



Lab-On-Time™ Strep A Rapid Test Device (Throat Swab)

BMT 51084 - Package Insert

An integrated rapid test device for throat swab specimen collection and qualitative detection of Strep A antigens. **For professional point of care use only.**

【INTENDED USE】

The Lab-On-Time™ Strep A Rapid Test Device is an integrated device for the collection of throat swab specimen and its rapid analysis by a chromatographic immunoassay for the qualitative detection of Strep A antigen to aid in the diagnosis of Group A Streptococcal infection.

【SUMMARY】

Streptococcus pyogenes is non-motile gram-positive cocci, which contains the Lancefield group A antigens that can cause serious infections such as pharyngitis, respiratory infection, impetigo, endocarditis, meningitis, puerperal sepsis, and arthritis.¹ Left untreated, these infections can lead to serious complications, including rheumatic fever and peritonsillar abscess.² Traditional identification procedures for Group A Streptococci infection involve the isolation and identification of viable organisms using techniques that require 24 to 48 hours or longer.^{3,4} The Lab-On-Time™ Strep A Rapid Test Device is an integrated device for throat swab sample collection and qualitative rapid analysis to detect the presence of Strep A antigen, providing results within 5 minutes. The test utilizes antibodies specific for whole cell Lancefield Group A Streptococcus to selectively detect Strep A antigens in a throat swab specimen.

【PRINCIPLE】

The Strep Lab-On-Time™ Strep A Rapid Test Device is an integrated device employing a qualitative, lateral flow immunoassay for the detection of Strep A carbohydrate antigen in a throat swab. In this test, antibody specific to Strep A carbohydrate antigen is coated on the test line region of the test. During testing, the extracted throat swab specimen reacts with an antibody to Strep A that is coated onto particles. The mixture migrates by capillary forces along the membrane to react with the antibody to Strep A on the membrane and generate a color line in the test line region. The presence of this color line in the test line region indicates a positive result, while its absence indicates a negative result. To serve as a procedural control, a colored line will always appear in the control line region, indicating that proper volume of specimen has been added and membrane wicking has occurred.

【REAGENT】

The test contains Strep A antibody coated particles and Strep A antibodies coated on the membrane.

【PRECAUTIONS】

- For professional in vitro diagnostic use only. Do not use after the expiration date.
- Do not eat, drink or smoke in the area where the specimens and kits are handled.
- Handle all specimens as if they contain infectious agents. Observe established precautions against microbiological hazards throughout the procedure and follow the standard procedures for proper disposal of specimens.
- Wear protective clothing such as laboratory coats, disposable gloves and eye protection when specimens are assayed.
- The used test should be discarded according to local regulations.
- Humidity and temperature can adversely affect results.
- Do not use test if pouch is damaged.

【STORAGE AND STABILITY】

Store as packaged in the sealed pouch at room temperature or refrigerated (2-30°C). The test is stable through the expiration date printed on the sealed pouch. The test must remain in the sealed pouch until use. **DO NOT FREEZE.** Do not use beyond the expiration date.

【SPECIMEN COLLECTION AND PREPARATION】

- If the package has been refrigerated, remove and let stand at room temperature (15-30°C) to allow time for test components to equilibrate prior to taking the throat swab for testing.
- Collect the throat swab specimen with the sterile *Strep Lab-On-Time™ Strep A Rapid Test Device* provided in the kit. Remove the device cap and use the device swab tip to swab the posterior pharynx, tonsils and other inflamed areas. Avoid touching the tongue, cheeks and teeth with the swab tip.⁵
- Testing should be performed immediately after the specimens have been collected.

【MATERIALS】

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|-----------------------------|---|
| Materials Provided | |
| • Strep A Rapid Test Device | • Extraction tube with Extraction Reagent (1.2M Sodium Nitrite) |
| • Package insert | • Workstation |

Materials Required But Not Provided

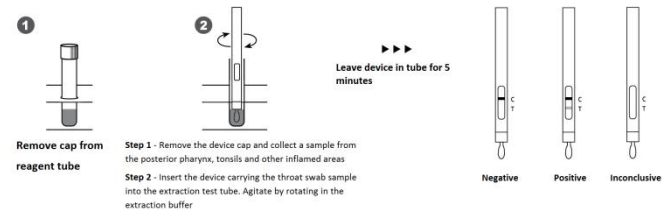
- Timer

【DIRECTIONS FOR USE】

Allow the test components and reagents to reach room temperature (15-30°C) prior to testing.

- Remove the Strep A Rapid Test Device from the sealed foil pouch and use it as soon as possible.

- Best results will be obtained if the test is performed immediately after opening the foil pouch.
- Remove the Extraction tube containing Extraction Reagent from the second sealed foil pouch and open the cap.
 - Insert the opened extraction tube upright in the round opening present on the backside of the box used to package the test.
 - Collect a throat swab from the patient's throat by swabbing the posterior pharynx, tonsils and other inflamed areas with the device's swab tip. Avoid touching the tongue, cheeks and teeth with the swab tip.
 - Insert the Strep A Rapid Test Device carrying the patient throat swab sample into the extraction test tube with the swab tip being dipped in the extraction buffer.
 - Agitate the Strep A Rapid Test Device by rotating it at least 10 times in the extraction tube for effective Strep A antigen extraction. Leave the Strep A Rapid Test Device in the extraction test tube and wait for the colored line(s) to appear. **Read the result in the Strep A Rapid Test Device results window between 5 and 10 minutes.** Do not interpret the result after 10 minutes.



【INTERPRETATION OF RESULTS】

(Please refer to the illustration above)

POSITIVE: Two lines appear* One colored line should be in the control line region (C) and another apparent colored line should be in the test line region (T). A positive result indicates that Strep A was detected in the specimen.

***NOTE:** The intensity of the color in the test line region (T) will vary depending on the concentration of Strep A present in the specimen. Therefore, any shade of color in the test line region (T) should be considered positive.

NEGATIVE: One colored line appears in the control line region (C). No line appears in the test line region (T). A negative result indicates that Strep A antigen is not present in the specimen or is present below the detectable level of the test. If clinical symptoms are not consistent with results, obtain another specimen for culture.

INVALID: Control line fails to appear. Insufficient specimen volume or incorrect procedural techniques are the most likely reasons for control line failure. Review the procedure and repeat the test with a new test. If the problem persists, discontinue using the test kit immediately and contact your local distributor.

【LIMITATIONS】

- The Strep A Rapid Test Device is for in vitro diagnostic use only. The test should be used for the detection of Strep A antigen in throat swab specimens only. Neither the quantitative value nor the rate of increase in Strep A antigen concentration can be determined by this qualitative test.
- This test will only indicate the presence of Strep A antigen in the specimen from both viable and non-viable Group A Streptococcus bacteria.
- A negative result should be confirmed by culture. A negative result may be obtained if the concentration of the Strep A antigen present in the throat swab is not adequate or is below the detectable level of the test.
- Excess blood or mucus on the swab specimen may interfere with test performance and may yield a false positive result. Avoid touching the tongue, cheeks, and teeth⁵ and any bleeding areas of the mouth with the Strep A Rapid Test Device when collecting specimens.
- As with all diagnostic tests, all results must be interpreted together with other clinical information available to the physician.

【PERFORMANCE CHARACTERISTICS】

Sensitivity and Specificity

Using three medical centers for evaluation, a total of 526 throat swabs were collected from patients exhibiting symptoms of pharyngitis. Each swab was rolled onto a sheep blood agar plate, and then tested by the Strep A Rapid Test Strip (Throat Swab). The plates were further streaked for isolation, and then incubated at 37°C with 5-10% CO₂ and a Bacitracin disk for 18-24 hours. The negative culture plates were incubated for an additional 18-24 hours. Possible GAS colonies were sub-cultured and confirmed with a commercially available latex agglutination grouping kit. Of the 526 total specimens, 404 were confirmed to be negative and 122 were confirmed to be positive by culture.

Method	Culture			Total Results
	Results	Positive	Negative	
Strep A Rapid Test Strip	Positive	116	9	125
	Negative	6	395	401
Total Results		122	404	526

Relative Sensitivity: 95.1% (89.6%-98.2%)* Relative Specificity: 97.8% (95.8%-99.0%)*
Overall accuracy: 97.1% (95.3%-98.4%)* * 95% Confidence Intervals

Positive Culture Classification	Strep A Rapid Test/Culture	% Agreement
Rare	8/10	80.0%
1+	18/20	90.0%
2+	19/20	95.0%
3+	33/34	97.1%
4+	38/38	100.0%

Cross Reactivity

The following organisms were tested at 2.5 x 10⁷ organisms per test and were all found to be negative when tested with the Strep A Rapid Test Strip. No mucoid-producing strains were tested.

Group B Streptococcus	Neisseria meningitidis	Serratia marcescens
Group F Streptococcus	Neisseria sicca	Klebsiella pneumoniae
Streptococcus pneumoniae	Branhamella catarrhalis	Bordetella pertussis
Streptococcus mutans	Group C Streptococcus	Neisseria gonorrhea
Staphylococcus aureus	Group G Streptococcus	Neisseria subflava
Corynebacterium diphtheria	Streptococcus sanguis	Hemophilus influenza
Candida albicans	Staphylococcus epidermidis	Pseudomonas aeruginosa
Enterococcus faecalis		

Analytical Sensitivity

The analytical performance of the device was determined by recording the visual results obtained using inactivated Strep A bacteria (strains ATCC #19615 and ATCC #12365) suspensions as positive control. The cutoff value (over 95% positive) was obtained at a bacterial load of **1*10⁵ organisms**.

Interfering Substances

The following table lists common, potentially interfering substances that have been tested and shown to not interfere with test performance and results:

Halls Cherry Cough Drops	Dimetapp Cough Syrup	Listerine Mouthwash
Halls Menthol Cough Drops	Vicks Chloraseptic Spray	Scope Mouthwash
Robitussin Cough Syrup	Cepacol Chloraseptic Spray	

【BIBLIOGRAPHY】

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	Attention, see instructions for use		Tests per kit		Authorized Representative
	For in vitro diagnostic use only		Use by		Do not reuse
	Store between 2-30°C		Lot Number		Catalog #
	Do not use if package is damaged				

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